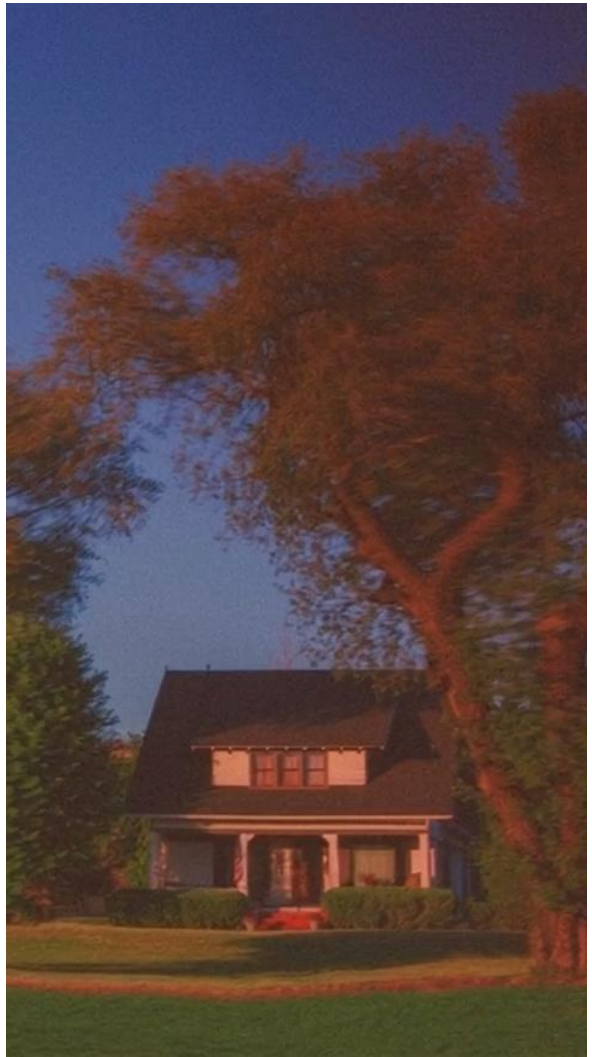
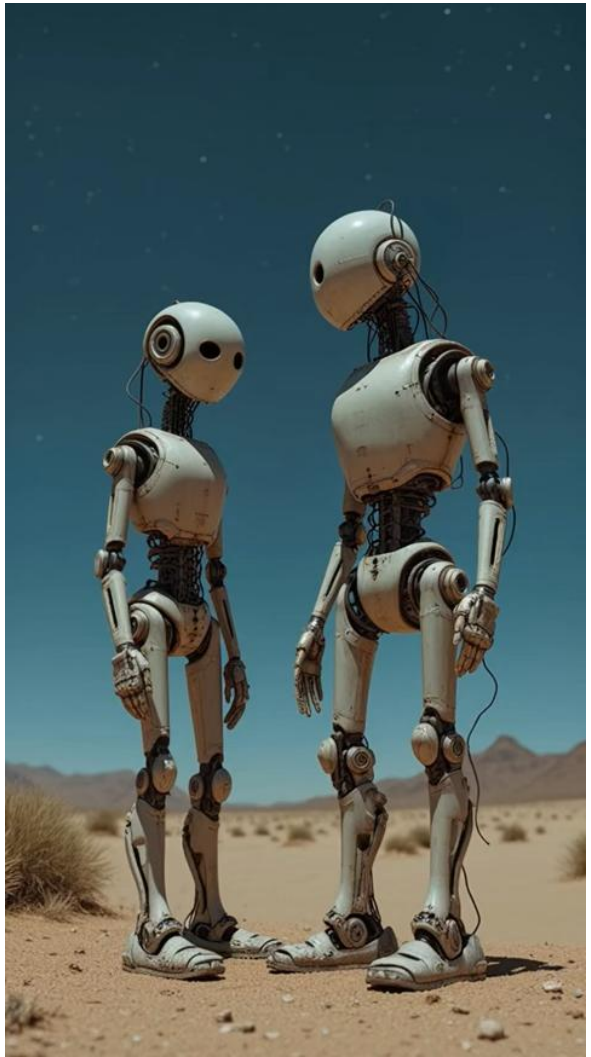
The goal of computer vision

- Visual content generation.

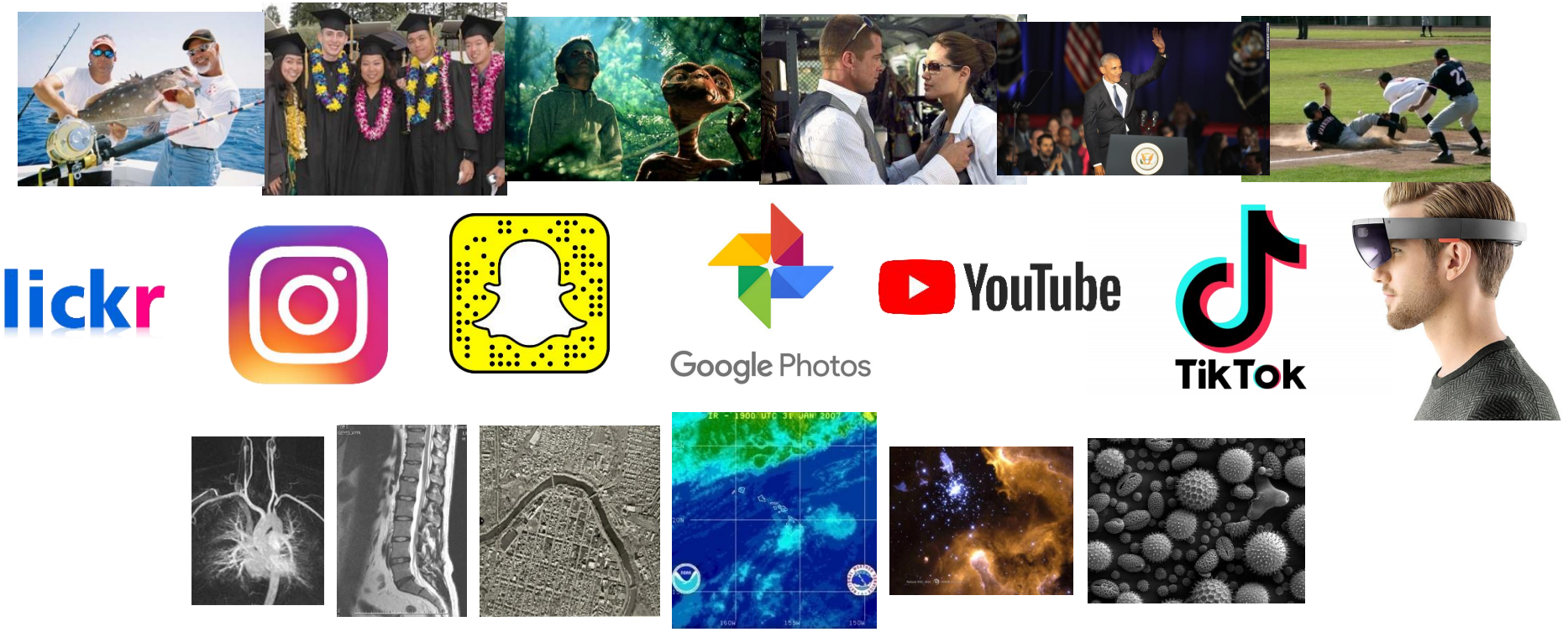





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Why study computer vision?

- Billions of images/videos captured per day



- Huge number of potential applications
- The next slides show the current state of the art

Optical character recognition (OCR)

- If you have a scanner, it probably came with OCR software



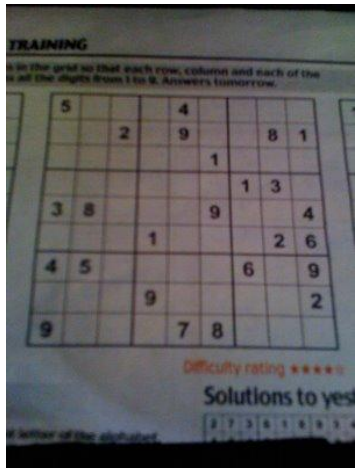
Digit recognition, AT&T labs (1990's)
http://yann.lecun.com/exdb/lenet/



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



Automatic check processing



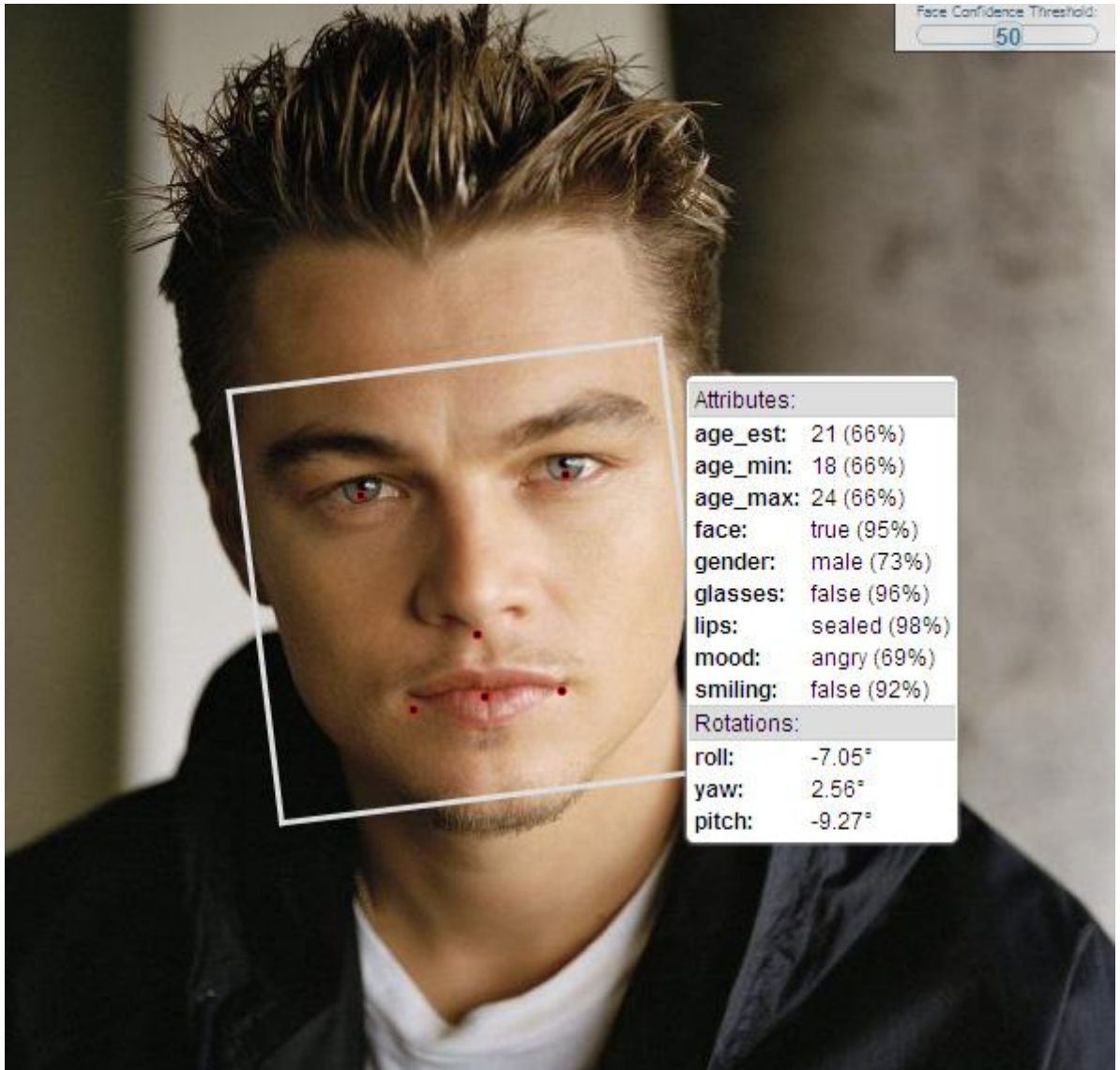
Sudoku grabber
http://sudokugrab.blogspot.com/

Face detection

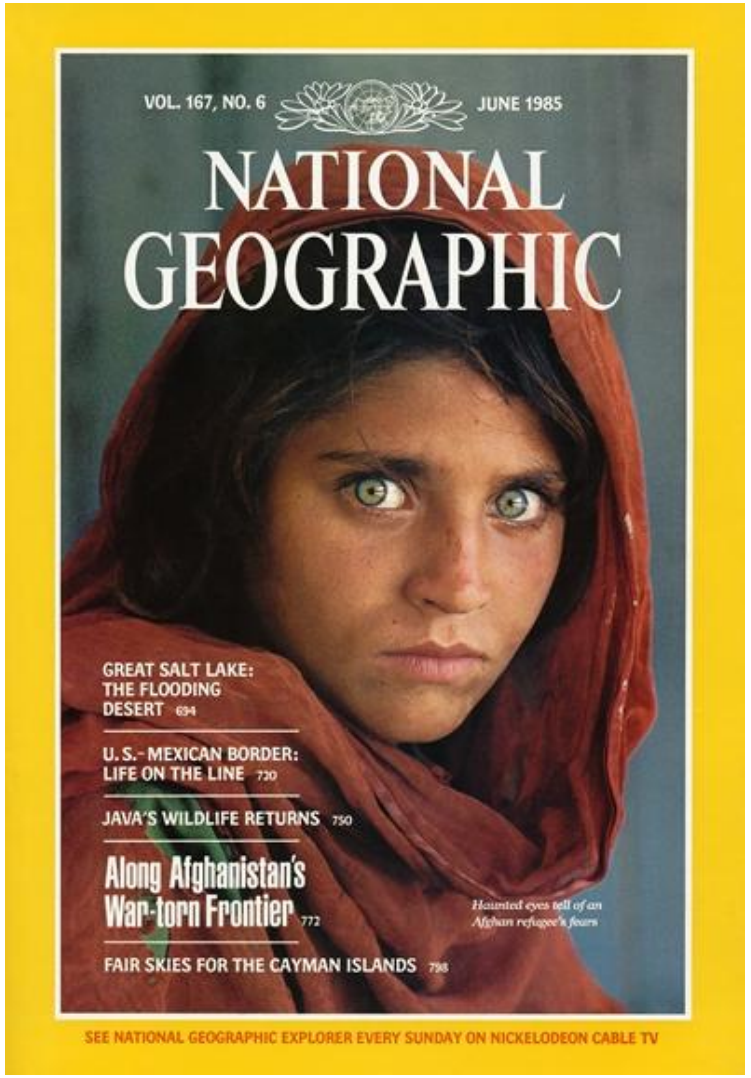


- Nearly all cameras detect faces in real time
  - (Why?)

Face analysis and recognition



Vision-based biometrics



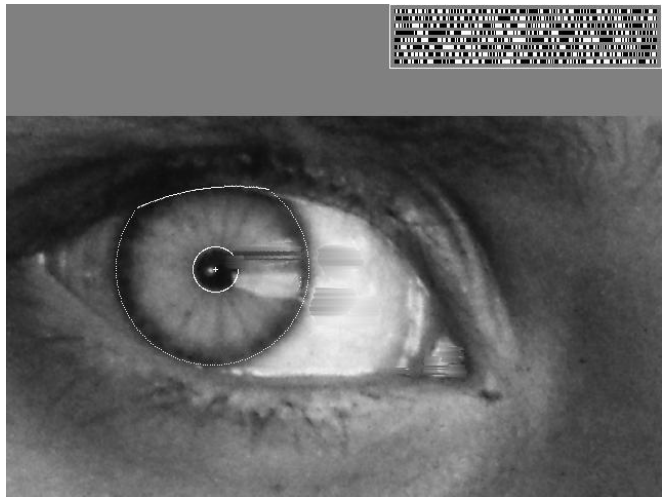
Who is she?

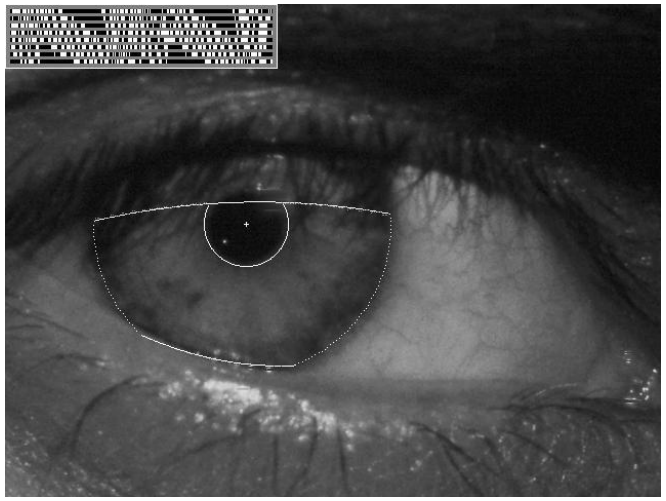
Source: S. Seitz

Vision-based biometrics



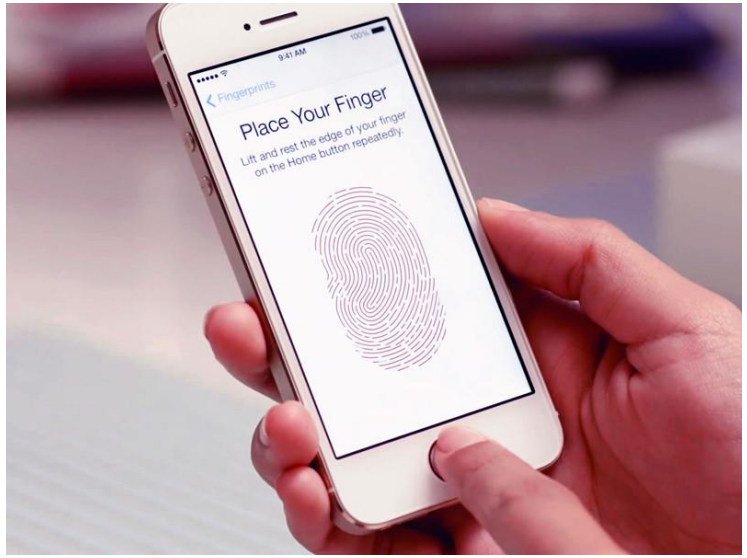
"How the Afghan Girl was Identified by Her Iris Patterns" Read the story





Source: S. Seitz

Login without a password



Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone X
See also http://www.sensiblevision.com/